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Cource Code : ITA 0329
Cource Name : Mobile Computing

CIA - 1

1. Given,

Total Cellular network = 1,200 km^2

radius = 1.8

Minimum Avaliable channel = 48 MHz

Total Spectrum = 200KHz

N = 7, 4

minimum reuse ratio = 4.5

reduction = 15%.

Number of

$$\text{Area} = \frac{3\sqrt{3}}{2} R^2 = 2.598(1.8)^2$$

$$= 2.6 \times 3.24 = 8.418 \ km^2$$

$$= 8.32$$

i) Total NO. of cell = $\frac{1200}{8.32} = 143.$

ii) Total aviolable channels(C) = $C = \frac{48 \times 1000}{200}$

$= 240$ Hz

iii) For N(7) = $\frac{240}{7} = 34.27$

For N(4) = $\frac{240}{4} = 60$

at N=7 reduction = 34.27×0.85

= 29 channels/unit

at N=4 = 60×0.85

= 51 channels/unit

iv) frequency reuse ratio for N = 7

$Q = \frac{D}{R} = \sqrt{3N}$

$R = \sqrt{3N} = \sqrt{21}$

distance

$D = Q \times R$

$= 4.5 \times \sqrt{21}$

$= 4.5 \times 4.1$

$= 8.25$

$= 4.58 \times 1.8$

$D = 8.28$

Frequency reuse ratio for N = 4

$R = \sqrt{3 \times 4} = \sqrt{12}$

$D = Q \times R$

$= 4.5 \times \sqrt{12}$

$= 4.5 \times 3.46$

$D = 6.24$

Therefore N = 7 is the usable choice because it provide better interference. and N = 4 is not sutable. The reuse distance at N = 7 is 8.28 and for N = 4 is 6.24.

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2. Given.

Cell boundary = 15.

inter PLMN transition = 2

inter PLMN handoff = 13

inter PLMN for BSS delay = 120ms

for NSS delay = 80ms.

additional inter PLMN delay = 350.

rooming register delay = 250

Potential patient loss = 512KPS

interuption limit = 3 sec

$T_h = T_{Bss} + N_{ss}$

$= 120 + 80 = 200s$

$D_{Total} = 200 \times 15 = 3000 ms$

$= 3 sec$

Total Normal hand off delay is 3sec

additional delay

Recovery $= 400 \times 3 = 1200 ms$

$= 1200 ms = 1.2 sec$

additional recovery delay = 1.2 sec

$D_{roming} = 350 + 250 = 600 ms$

$= 600 \times 2 = 12000 ms$

roaming delay $= 1.2 sec$

Overall Communication intruption.

$D_{Total} = 3 + 1.2 + 1.2 sec$

$= 5.4 sec$ or $5400 ms$

System satisfies the interuption limit is 3sec

$5.4 sec > 3 sec$

Time is exceed the time limit is 2sec

Potential data loss:

$D_{data} = 512 kbps$

$= 512,000 bits$

$T_{total} = \frac{512,000}{8 \times 1000} = 64.$

» Given,

Total transceiver stations (BTS) : 12.

TRx Per sector : 3 sector

TRX Per transceiver = 4

TDM Per sector = 8 timeslots

timeslots Per sector = 2

call per hour = 1.2

Subscribers = 1.2000

avg Call holding time = 90 sec

Abs interface = 16 Kbps

E1 link capacity = 2048

max capacity = 80%.

i) TCH = (TRX x timeslots) - Signal by time sols

= 4 x 8 - 2

= 30 time slots

ii) Traffic Per sector = 30

Per BTS = 30(3) = 90

BSC = 90(12) = 1080.

b. Offered traffic Per Sector

$$A = \frac{\text{Calls Per hour} \times \text{Call during in sec}}{3600}$$

$$= \frac{1200 \times 1.2 \times 90}{3600}$$

A = 36 range.

c. Channel Utilization

$$U = \frac{\text{Offered traffic}}{\text{Available channels}} \times 100.$$

$U = \frac{36}{30} \times 100 = 120\%$

$U = 120\%$

d. Minimum TRXs required per sec.

$A_{max} = 0.8(8T-2)$

No. of TRXs per sector =

$36 \leq 0.8(8T-2)$

$45 \leq 8T-2$

$47 \leq 8T$

$T \geq 5.875$

$\therefore T = 6$ TRXs per sec

Utilization $= \frac{36}{46} \times 100 \Rightarrow 78\%$

3ii) Given:

Cell-boundary Crossi Crossing = 15MHz

BSS Corrdination delay = 120

inter PLMN transition = 2

Authentication delay = 350

Roaming registration = 250ms Data rate = 9.6 kbps

Failed handoffs = 3 Chip rate = 1.2288 MCPS

Recbr $E_b/N_D = 8$

Voice activity factor = 0.4

frequency efficiency = 0.6

CDMA reuse factor = 1

i) FDMA Capacity

$= 10000/30$

$= 333.33$

per cell

$= 333/7$

$= 47$ channels

ii) TDMA Capacity
= 10000/200
= 50 Capacity

Total :
= 50 × 8
= 400 channels

Channel Percell
= 400/4
= 100 channels

iii) CDMA processing gain
$P_G = W/R$
= 1228.8/9.6
= 128

In decibles
$PG = 10\log(128)$
= 21.07 db

1288/96

iv) CDMK 1.5 per Carrier
$K = \frac{GP}{(Eb/No)} = \frac{128(0.6)}{8(0.4)}$

Percell = 8 (24) = 192

192 user percell.

v) Capacity Compression Techniques
FDMA,
TDMA.
CDMA.

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4. Given.
Total imaging file = 240.
Each size = 18MS.
Allocated time slots = 4.
Congestive limit effective = 20%.
data rate = 20 kbps
each lasting = 2.5 s
UMTS transfer = 384 Kbps
emergency da to delivery = 12 h

i) Aggregrate GPRS rate

$= 4 \times 20$

$= 80 \text{ Kbps}$

ii) Effective GPRS throughput

$= 80 \times (1 - 0.18) \times 0.85$

$= 80 \times 0.82 \times 0.85$

$= 55.76$

iii) Original data volume

$= 240 \times 18$

$= 4320 \text{ Mb}$

$= 4320 \times 8$

$= 34560$

iv) Data retransimission

$= 34560 \times (1 + 0.12)$

$= 38707.2 \text{ Mb}$